JavaScript Object Accessors

Previous Next

JavaScript Accessors (Getters and Setters)

Getters and setters allow you to define Object Accessors (Computed Properties).

JavaScript Getter (The get Keyword)

This example uses a `lang` property to `get` the value of the `language` property.

Example

```
// Create an object:
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "en",
  get lang() {
    return this.language;
  }
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.lang;
```

Try it Yourself »

JavaScript Setter (The set Keyword)

```
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "",
  set lang(lang) {
    this.language = lang;
  }
};

// Set an object property using a setter:
person.lang = "en";

// Display data from the object:
document.getElementById("demo").innerHTML = person.language;
```

Try it Yourself »

ADVERTISEMENT

JavaScript Function or Getter?

What is the differences between these two examples?

Example 1

```
const person = {
  firstName: "John",
  lastName: "Doe",
```

```
// Display data from the object using a method:
document.getElementById("demo").innerHTML = person.fullName();
```

Try it Yourself »

Example 2

```
const person = {
  firstName: "John",
  lastName: "Doe",
  get fullName() {
    return this.firstName + " " + this.lastName;
  }
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.fullName;
```

Try it Yourself »

Example 1 access fullName as a function: person.fullName().

Example 2 access fullName as a property: person.fullName.

The second example provides a simpler syntax.

Data Quality

JavaScript can secure better data quality when using getters and setters.

Using the `lang` property, in this example, returns the value of the `language` property in upper case:

Example

```
// Create an object:
const person = {
  firstName: "John",
  lastName: "Doe",
```

```
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.lang;
```

Try it Yourself »

Using the `lang` property, in this example, stores an upper case value in the `language` property:

Example

```
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "",
  set lang(lang) {
    this.language = lang.toUpperCase();
  }
};

// Set an object property using a setter:
person.lang = "en";

// Display data from the object:
document.getElementById("demo").innerHTML = person.language;
```

Try it Yourself »

Why Using Getters and Setters?

- It gives simpler syntax
- It allows equal syntax for properties and methods
- It can secure better data quality
- It is useful for doing things behind-the-scenes

Object.defineProperty()

The `Object.defineProperty()` method can also be used to add Getters and Setters:

```
// Define object
const obj = {counter : 0};

// Define setters and getters
Object.defineProperty(obj, "reset", {
  get : function () {this.counter = 0;}
});
Object.defineProperty(obj, "increment", {
  get : function () {this.counter++;}
});
Object.defineProperty(obj, "decrement", {
  get : function () {this.counter--;}
});
Object.defineProperty(obj, "add", {
  set : function (value) {this.counter += value;}
});
Object.defineProperty(obj, "subtract", {
  set : function (value) {this.counter -= value;}
});

// Play with the counter:
obj.reset;
obj.add = 5;
obj.subtract = 1;
obj.increment;
obj.decrement;
```

Try it Yourself »

Learn More:

JavaScript Object Definitions

JavaScript Object Properties

JavaScript Object Methods

JavaScript Object Display

JavaScript Object Constructors

JavaScript Object this

JavaScript Object Iterations

JavaScript Object Management

JavaScript Object Getters & Setters

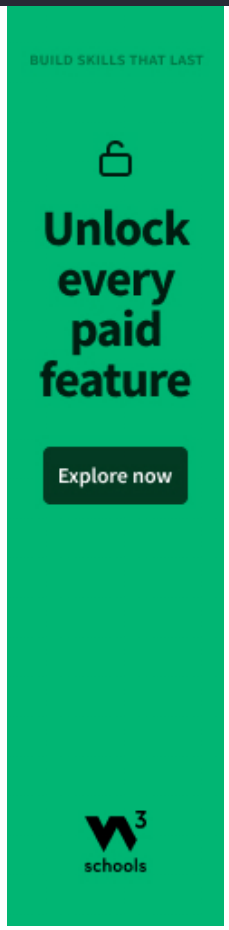
JavaScript Object Protection

JavaScript Object Reference

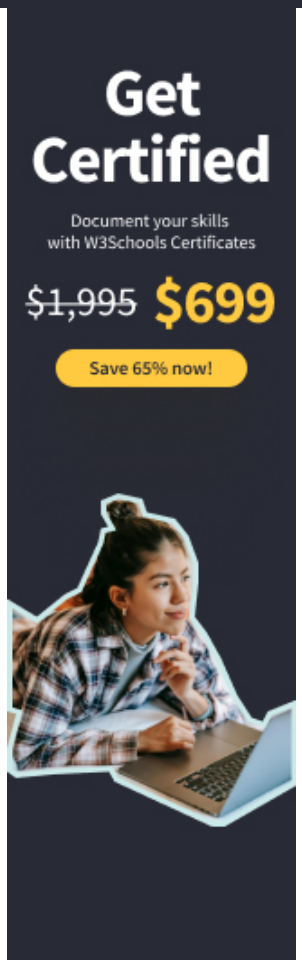
❮ Previous
Next ❯

Track your progress - it's free!
Sign Up
Log in

ADVERTISEMENT



COLOR PICKER



ADVERTISEMENT

ADVERTISEMENT

Top Tutorials

HTML Tutorial
CSS Tutorial
JavaScript Tutorial
How To Tutorial
SQL Tutorial
Python Tutorial
W3.CSS Tutorial
Bootstrap Tutorial
PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

Top References

HTML Reference
CSS Reference
JavaScript Reference
SQL Reference
Python Reference
W3.CSS Reference
Bootstrap Reference
PHP Reference
HTML Colors
Java Reference
Angular Reference
jQuery Reference

Top Examples

HTML Examples
CSS Examples

Get Certified

HTML Certificate
CSS Certificate

W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookie and privacy policy.

Copyright 1999-2025 by Refsnes Data. All Rights Reserved. W3Schools is Powered by W3.CSS.